

Prakhar

Research Intern

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SUMMARY

PhD student in Computer Science at UMBC specializing in Machine Learning, Reinforcement Learning, Mathematical Reasoning in LLMs and Agentic memory and Long-term Memory. My Research focuses on **enhancing mathematical reasoning** in LLMs, improving RL sample efficiency, and building **smarter memory** systems for **Agentic AI workflows** using human-driven techniques.

EXPERIENCE

Applied Scientist Intern

Microsoft

May '26 — Aug '26
Redmond, WA, United States

- Worked with the **Microsoft Copilot Studios(MCS)** Team to improve and optimise the agent skills inside the MCS platform.
- Developed a standalone evaluation scoring system to access the performance of the Agent Skills.

Summer Research Intern

Emergence AI

May '25 — Oct '25
New York City, United States

- Worked with a long-term memory system for conversational agents.
- Achieved state-of-the-art (SOTA) results on the **LongMemEval** benchmark with an accuracy of **86%**, surpassing the best SOTA by **15%**
- Developed specialised agents for diverse enterprise tasks, particularly in planning and self-improvement.

Graduate Research Assistant

University of Maryland Baltimore County

Jan '22 — Present
Baltimore, United States

- Currently working on enhancing mathematical reasoning capabilities in large language models through human-driven learning techniques.
- Designed a Sim2Real framework, training RL algorithms with YOLO object detectors in simulation, achieving an **85%** transfer success to real-world robots.
- Worked on Reinforcement Learning using Human Feedback(RLHF) technique TAMER (Scalar binary feedback) and showcased how can agents harness the information contained in human-generated signals of reward to learn sequential decision-making tasks.

Graduate Assistant

University of Maryland Baltimore County

Aug '21 — Dec '21
Baltimore, United States

- Assisted in teaching CMSC 691.4 (Introduction to Data Science), contributing to the education of over 50 students by grading assignments and providing feedback, ensuring a 95% assignment completion rate.
- Collaborated with a professor to implement student projects, increasing student engagement and project completion rates by 20%.

Software Engineer

Titan Company Limited

Jul '19 — Apr '21
Bengaluru, India

- Developed a web application using Spring Boot and REST APIs, enabling over 8,000 employees to efficiently visualize and manage their asset data stored in Oracle DB, resulting in a 40% reduction in data retrieval time.
- Implemented a comprehensive Python dashboard to analyze daily login data, enabling the derivation of actionable insights and significantly improving the accuracy of user engagement tracking by 25%.

Software Engineer Intern

Titan Company Limited

Mar '19 — Jun '19
Bengaluru, India

- Participated in testing and improving various applications within the software development cycle, enhancing software reliability by 30% and reducing bug occurrences by 20%.
- Tested and managed BPM-related applications, increasing process efficiency by 25% and ensuring seamless workflow integration.

Research Intern

National University of Singapore

Jun '18 — Jul '18
Singapore City, Singapore

- Examined a Credit card fraud detection dataset achieving an accuracy of **98.5 %** using logistic regression, which outperformed ANN, random forest, decision tree, and Support vector machines (SVM)
- Generated advanced data preprocessing, reducing data redundancy by 25% and model training time by 30%, boosting efficiency.

EDUCATION

PhD in Computer Science, University Of Maryland Baltimore County (GPA: 3.589)

Aug '23 — Present
Baltimore, United States

- Working to find out how the math reasoning is so poor in LLMs

MS in Computer Science, University Of Maryland Baltimore County (GPA: 3.5)

Aug '21 — May '23
Baltimore, United States

Thesis - Dynamic Edge Weighting in Relational Graph Convolutional Networks: Enhancing Sample Efficiency via Graph Attention in Reinforcement Learning	
B.Tech in Computer Science Engineering , SRM Institute of Science and Technology (GPA: 3.7)	Aug '15 — May '19 Chennai, India
Final Year Project - Neural Sketching	
AWARDS	
Winner of Mega Tetris, Human-Guided System Adaptation (HSA) Science Hackathon	Aug '22
Columbia University & DEVCOM Army Research Labs	
Winner of the Human-Guided System Adaptation (HSA) Science Hackathon by DEVCOM Army Research Laboratory in Collaboration with Columbia University	
Travel Grant Winner and Presenter at Pycon 2024	May '24
Python Software Foundation	
Awarded Travel grant to present Poster at World largest python Conference Pycon 2024 at Pittsburgh, Pennsylvania	
Presenter at AI and Autonomy for Multi-Agent Systems(ArtiAmas) Annual Review 2022	May '22
University of Maryland College Park	
Selected and presented my Research work in front of Researchers of United States Army Research Laboratory, United States Naval Research, Professors of University of Maryland College Park at Maryland Robotics Center in University of Maryland College Park.	
PROJECTS	
ScholarAgent Link	Feb '25
An AI Agent that fetches the recent ArXiv research papers based on search, just like Google Scholar but in a more user-friendly and readable way	
LLM Detect AI Text Link	Jan '24
Developed a machine learning model that can accurately detect whether an essay was written by a student or an LLM, finishing in the top 25% of leaderboard.	
Deep Learning Steering Model Link	Aug '17
Built a self-driving prototype using Convolutional Neural Networks and OpenCV. The system used images and steering angles as labels to determine the correct steering angle for the vehicle.	
POSTERS AND PRESENTATIONS	
- Presentation @ 3rd WDT Semi Annual Review 2025 held at University of Maryland, Baltimore County - Poster Presentation at PyCon 2023 Link - Presentation @ AI and Autonomy for Multi-Agent Systems(ArtiAmas) Annual Review 2022 at University of Maryland College Park Link - Poster Presentation at PyCon 2024 Link	
PROFESSIONAL SERVICE	
Hackathon Judge , VTHacks	Blacksburg, VA Sep '26
Judge for VTHacks 14 Virginia Tech's Flagship Hackathon , Judged more than 160 teams and 500+ participants.	
Reviewer , ICML 2026	South Korea May '26
Paper Reviewer for "AI4Math @ ICML 2026"	
Reviewer , NeurIPS 2025	San Diego Oct '25
Reviewer for Workshop on "Mathematical Reasoning and AI MATH-AI@ NeurIPS'25"	
Reviewer , NeurIPS 2026	Atlanta, Georgia Sep '26 — Dec '26
Reviewer for "The 6th Workshop on Mathematical Reasoning and AI" @ NeurIPS 2026	
SKILLS	
Programming Python, Java, SQL, JavaScript, C++	
Machine Learning Deep Reinforcement Learning(DRL), Logistic Regression, Neural Networks (ANN, CNN), YOLO Object Detection, Sim2Real, Large Language Models(LLMs), Retrieval Augmented Generation(RAG)	
Database Management MySQL, Oracle DB, CRUD operations	
Software Development Spring Boot, REST APIs, Android Studio, Gradle, MERN stack (MongoDB, Express.js, React, Node.js)	
PUBLICATIONS	
ISM: Self-Improving Strategy Memory for Continual Mathematical Reasoning	Jun '26
ICML 2026	
Paper Link	
Honest Lying: Understanding Memory Confabulation in Reflexive Agents	Jun '26
ICML 2026	
Paper Link	

WildfireVLM: AI-powered Analysis for Early Wildfire Detection and Risk Assessment Using Satellite Imagery IGARSS 2026 Paper Link	Mar '26
SOTA Results in Agentic Memory on LongMemEval Emergence AI Blog Link	Jul '25
SBI-RAG: Enhancing Math Word Problem Solving for Students through Schema-Based Instruction and Retrieval Augmented Generation NeurIPS 2024 Paper Link	Dec '24
Dynamic Edge Weighting in Relational Graph Convolutional Networks: Enhancing Sample Efficiency via Graph Attention in Reinforcement Learning University of Maryland, Baltimore County ProQuest Dissertations & Theses Thesis Link	Sep '24
ReProHRL: Towards Multi-Goal Navigation in the Real World using Hierarchical Agents AAAI 2023 Paper Link	Feb '23
Toward Real-World Implementation of Deep Reinforcement Learning for Vision-Based Autonomous Drone Navigation with Mission RSS 2022 Paper Link	Jul '22

PEER-REVIEWING EXPERIENCE

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- Reviewer for Workshop on Mathematical Reasoning and AI MATH-AI @ NeurIPS'25.
 - Reviewer for **Inductive Biases in Reinforcement Learning Workshop @ Reinforcement Learning Conference 2025(RLC 2025)**
 - Reviewed 3 paper for **Association for Computational Linguistics (ACL 2025)**
 - Talks and Tutorial Reviewer for **Python Conference(Pycon 2025)**
 - Reviewer for 3rd AI for Math Workshop @ ICML 2026